

2.2 Logarithmic and exponential functions

Pure Mathematics 2 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

$\ln$ and $\exp$ are inverses on appropriate domains. Log rules combine products into sums and powers into coefficients; do not split $\ln(a+b)$. Exponential growth/decay solves $\frac{dy}{dt}=ky$ and requires the initial condition to determine the multiplier.

Worked example: follow the reasoning

$e^{(\ln 5)}=5$ and $\ln(e^3)=3$; $\ln(2x)=\ln 2+\ln x$ for $x>0$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Solve $\ln(x-1)=2$ for real x .

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Domain $x > 1$ [1]; exponentiate: $x-1=e^2$ [1]; $x=1+e^2$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.